

1

|          | Unvaccinated | Vaccinated | Total  |
|----------|--------------|------------|--------|
| Death    | 253          | 481        | 734    |
| Survived | 150799       | 116633     | 267432 |
| Total    | 151052       | 117114     | 268166 |

i

Odds\_Overall  $\frac{\text{\#Deaths of Vaccinated PPL}}{\text{\#PPL Survived w/ Vaccine}} = \frac{481}{116633}$

ii

Under 50

|          | Unvaccinated | Vaccinated | Total  |
|----------|--------------|------------|--------|
| Death    | 48           | 21         | 69     |
| Survived | 147564       | 89786      | 237350 |
| Total    | 147612       | 89807      | 237419 |

Odds\_Under50  $\frac{\text{\#Deaths of Vaccinated PPL}}{\text{\#PPL Survived w/ Vaccine}} = \frac{21}{89786}$

Over 50

|          | Unvaccinated | Vaccinated | Total |
|----------|--------------|------------|-------|
| Death    | 205          | 460        | 265   |
| Survived | 3235         | 26847      | 30082 |
| Total    | 3440         | 27307      | 30747 |

Odds\_Over50  $\frac{\text{\#Deaths of Vaccinated PPL}}{\text{\#PPL Survived w/ Vaccine}} = \frac{460}{26847}$

iii

$$\text{Odds\_Overall} = \frac{\# \text{Deaths of Vaccinated PPL}}{\# \text{PPL Survived w/ Vaccine}} = \frac{253}{150799}$$

$$\text{OR\_Overall} = \frac{253 * 116633}{150799 * 481} = 0.4068163789$$

$$\text{Odds\_Under50} = \frac{\# \text{Deaths of Vaccinated PPL}}{\# \text{PPL Survived w/ Vaccine}} = \frac{48}{147564}$$

$$\text{OR\_Under50} = \frac{48 * 89786}{147564 * 21} = 1.3907534552$$

$$\text{Odds\_Over50} = \frac{\# \text{Deaths of Vaccinated PPL}}{\# \text{PPL Survived w/ Vaccine}} = \frac{205}{3235}$$

$$\text{OR\_Over50} = \frac{205 * 26847}{3235 * 460} = 3.6984308850$$

Here, all odds ratios are outside of [0.8, 1.2] which would suggest that the vaccination has a relevant change to the odds of a person dying to the disease. According to the data, vaccination decreased a person's odds of survival when looking at the data overall, however when we break the data down by it's strata, then we find that vaccination increases a person's odds of survival for each strata.

## iv

Yes simpson's paradox is occurring. It is clear in by the Odds Ratios calculated in the above problem that overall it appears that vaccination causes an increased risk of death but when calculating the Odds Ratios by each strata, it comes out that the vaccine increases a person's odds of survival for both stratas. Thus. simpson's paradox is occurring because the data shows different changes from the vaccine based on if you look at the overall statistic vs the statistics at teh strata level.

## 2

## i

```
diabetes = read.csv("data/diabetec.csv")
```

```
head(diabetes)
```

```
  age ratio gender  frame diabetic
1  46   3.6 female medium         0
2  29   6.9 female  large         1
3  58   6.2 female  large         0
4  67   6.5  male  large         0
5  64   8.9  male medium         0
6  34   3.6  male  large         0
```

```
model = glm(diabetic ~ age + gender + ratio + frame, data = diabetes, family = binomial(link="logit"))
summary(model)
```

```
summary(model)
```

Call:

```
glm(formula = diabetic ~ age + gender + ratio + frame, family = binomial(link = "logit"),
     data = diabetes)
```

Coefficients:

|             | Estimate  | Std. Error | z value | Pr(> z ) |     |
|-------------|-----------|------------|---------|----------|-----|
| (Intercept) | -2.746860 | 0.567417   | -4.841  | 1.29e-06 | *** |
| age         | 0.033862  | 0.007327   | 4.622   | 3.80e-06 | *** |
| gendermale  | -0.207254 | 0.231418   | -0.896  | 0.37047  |     |
| ratio       | 0.211458  | 0.071379   | 2.962   | 0.00305  | **  |
| framemedium | -0.363148 | 0.270488   | -1.343  | 0.17941  |     |
| framesmall  | -0.287879 | 0.322001   | -0.894  | 0.37130  |     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 517.77 on 389 degrees of freedom  
Residual deviance: 473.90 on 384 degrees of freedom  
AIC: 485.9

Number of Fisher Scoring iterations: 4

ii

```
reduced_model = glm(diabetic ~ age + ratio, data = diabetes, family = binomial(link="logit"))
anova(reduced_model, model, test="Chisq")
```

Analysis of Deviance Table

Model 1: diabetic ~ age + ratio

Model 2: diabetic ~ age + gender + ratio + frame

|   | Resid. Df | Resid. Dev | Df | Deviance | Pr(>Chi) |
|---|-----------|------------|----|----------|----------|
| 1 | 387       | 476.16     |    |          |          |
| 2 | 384       | 473.90     | 3  | 2.2528   | 0.5216   |

According to the nested model test, the extra predictors (gender and frame) do not have a significant association with the response variable diabetes.

iii

```
summary(reduced_model)
```

Call:

```
glm(formula = diabetic ~ age + ratio, family = binomial(link = "logit"),
     data = diabetes)
```

Coefficients:

|             | Estimate  | Std. Error | z value | Pr(> z ) |     |
|-------------|-----------|------------|---------|----------|-----|
| (Intercept) | -3.159264 | 0.469899   | -6.723  | 1.78e-11 | *** |
| age         | 0.035272  | 0.007167   | 4.921   | 8.59e-07 | *** |
| ratio       | 0.214832  | 0.069339   | 3.098   | 0.00195  | **  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 517.77 on 389 degrees of freedom  
 Residual deviance: 476.16 on 387 degrees of freedom  
 AIC: 482.16

Number of Fisher Scoring iterations: 4

Thus age has a coefficient of 0.035272 which represents the change in log odds of diabetes for every unit increase of age, while holding all other predictors constant.

Thus ratio has a coefficient of 0.214832 which represents the change in log odds of diabetes for every unit increase of ratio, while holding all other predictors constant.

## iv

```
library(ResourceSelection)
```

Warning: package 'ResourceSelection' was built under R version 4.4.3

ResourceSelection 0.3-6 2023-06-27

```
HL <- hoslem.test(model$y, reduced_model$fitted.values, g=7)
HL
```

Hosmer and Lemeshow goodness of fit (GOF) test

data: model\$y, reduced\_model\$fitted.values  
 X-squared = 6.1897, df = 5, p-value = 0.2882

This resulted in a p-value of 0.2882 which is greater than  $\alpha = 0.05$  so we do not reject the null hypothesis and the model is a good fit for the data.

V

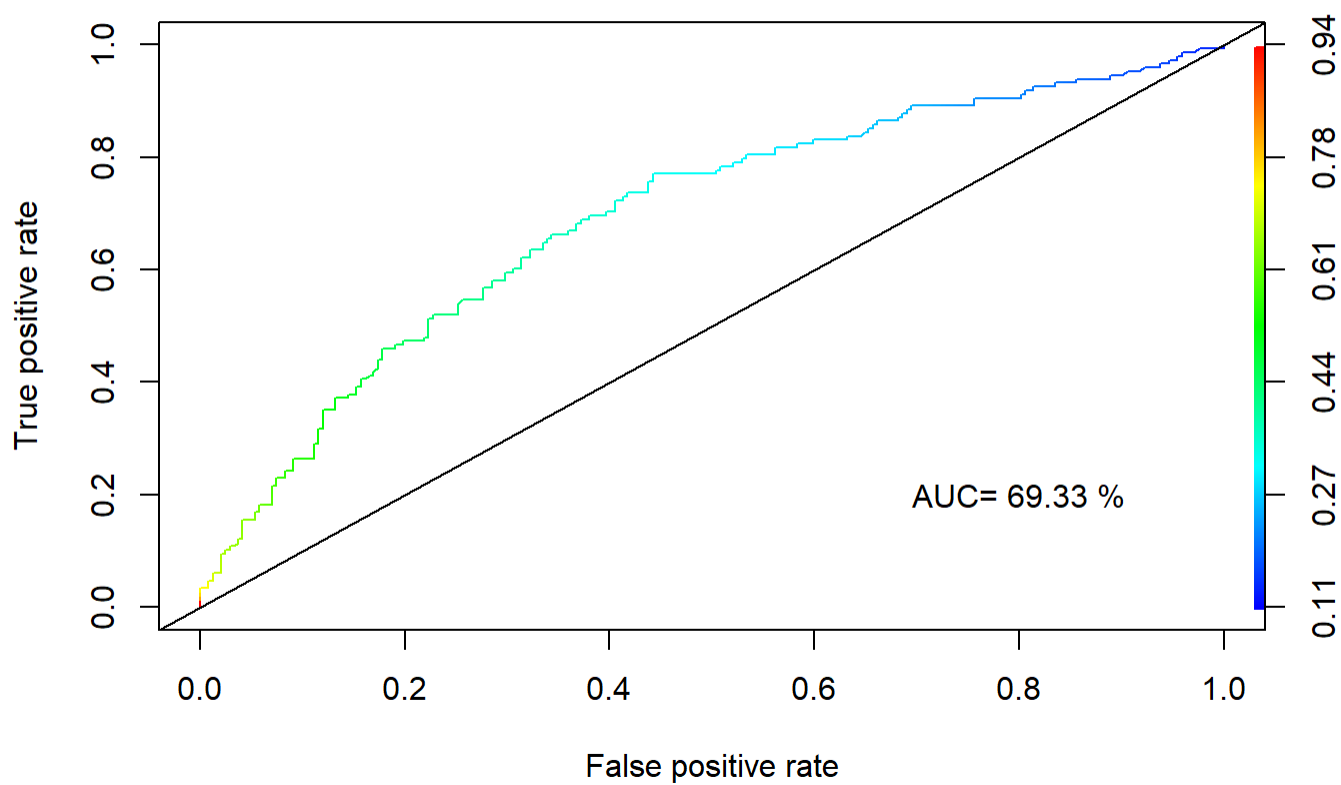
```
library(ROCR)
```

Warning: package 'ROCR' was built under R version 4.4.3

```
## ROC
pred <- prediction(reduced_model$fitted.values, diabetes$diabetic)
perf <- performance(pred, measure="tpr", x.measure = "fpr")

## area under curve
auc <- performance(pred, "auc")
auc <- round(as.numeric(auc@y.values),4)

plot(perf, colorize=T)
abline(a=0, b=1)
text(0.8,0.2, labels= paste("AUC=", auc*100, "%") )
```



vi

```
cost_perf = performance(pred, "cost")

optimal_pi0 <- pred@cutoffs[[1]][which.min(cost_perf@y.values[[1]])]
optimal_pi0
```

94

0.4630625

Thus the optimal cutoff is 0.4630625